

AI/ML-Based Predictive Maintenance and Vehicle Health Monitoring System

Abhishek Ghosh

Department of Computer Science / AI & ML, Institute of Engineering and Management, New Town
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Abstract—Predictive maintenance aims to identify abnormal vehicle conditions before a component failure occurs. This paper presents an AI/ML-based vehicle health monitoring system using operational parameters including engine temperature, RPM, oil pressure, vibration, battery voltage, coolant temperature, fuel consumption, vehicle speed, and operating hours. The workflow performs data preprocessing, train-test splitting, feature scaling, and classification using a Random Forest model. The trained model estimates failure risk and converts the prediction probability into a vehicle health score. Based on predefined thresholds, the system reports NORMAL, WARNING, or CRITICAL status and generates a maintenance alert. Feature-importance analysis is also used to identify influential vehicle parameters. The prototype demonstrates an end-to-end predictive-maintenance workflow and provides a foundation for future integration with authentic vehicle telemetry and real-time monitoring.

Keywords—Predictive Maintenance, Vehicle Health Monitoring, Artificial Intelligence, Machine Learning, Random Forest, Failure Prediction, Vehicle Health Score

I. INTRODUCTION

Vehicle failures can reduce safety, increase maintenance costs, and cause unexpected downtime. Traditional maintenance often depends on fixed service intervals or inspection after a problem becomes visible. A predictive maintenance system instead uses operating data to estimate the possibility of failure and provide an early warning.

The proposed architecture follows the sequence of vehicle sensors, data preprocessing, machine-learning model, failure probability, vehicle health score, and maintenance alert. This workflow is designed to demonstrate how machine learning can support vehicle condition monitoring and maintenance decision support.

II. PROJECT OBJECTIVES

1) Collect and organize vehicle operating parameters for health monitoring. 2) Preprocess the input data for machine-learning use. 3) Train a Random Forest classifier to identify failure risk. 4) Estimate failure probability for test samples. 5) Convert failure probability into a vehicle health score. 6) Classify vehicle condition as NORMAL, WARNING, or CRITICAL. 7) Generate a maintenance alert based on predicted condition. 8) Analyze feature importance. 9) Visualize health score, failure probability, and status distribution.

III. SYSTEM REQUIREMENTS

Programming language: Python. Development environment: Google Colab / Jupyter Notebook. Data format: CSV. Main machine-learning model: Random Forest Classifier. Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn. Target variable: Failure, where 0 represents no failure and 1 represents failure.

IV. INPUT PARAMETERS AND DATASET

The system uses nine vehicle operating parameters: engine temperature, RPM, oil pressure, vibration, battery voltage, coolant temperature, fuel consumption, vehicle speed, and operating hours. Each parameter represents a relevant aspect of engine, mechanical, electrical, cooling, fuel, speed, or accumulated operating condition.

For the implemented prototype, a CSV dataset named `vehicle_health_dataset.csv` was created for the nine specified parameters and a binary Failure target. The prototype contains 1,000 samples. The dataset demonstrates the complete machine-learning pipeline in Google Colab. It is a prototype dataset and should not be presented as real-world vehicle sensor data; real deployment requires

authentic vehicle telemetry and properly labelled failure events.

V. SYSTEM ARCHITECTURE

The proposed workflow is: Vehicle Sensors → Data Preprocessing → Machine Learning Model → Failure Probability → Vehicle Health Score → Maintenance Alert. Sensor inputs include RPM, temperature, pressure, and vibration along with the other defined operating parameters.

VI. DATA PREPROCESSING

The CSV dataset is loaded using Pandas. The Failure column is separated as the target variable y , while the remaining nine parameters form the input feature matrix X . The data is divided into training and testing sets using an 80:20 split. StandardScaler is then applied to scale the input features.

VII. RANDOM FOREST MODEL

Random Forest Classifier is selected as the main model for the implemented prototype. It combines multiple decision trees and uses their combined predictions for classification. The model receives the preprocessed vehicle parameters and predicts whether a failure condition is likely.

The implemented configuration uses 100 decision trees ($n_{\text{estimators}} = 100$) with $\text{random_state} = 42$ for reproducibility. The model is trained using the training portion of the dataset and evaluated on held-out test data.

VIII. FAILURE PROBABILITY AND VEHICLE HEALTH SCORE

After training, the model's `predict_proba()` function is used to obtain the estimated probability of the Failure = 1 class. The probability is converted into percentage form:

Failure Probability (%) = Predicted Failure Probability $\times$ 100

The prototype defines the vehicle health score as the complement of failure probability:

Vehicle Health (%) = 100 – Failure Probability (%)

IX. STATUS CLASSIFICATION AND MAINTENANCE ALERT

The health score is mapped to three project-defined conditions. A score of 70% or above is classified as NORMAL and indicates that no maintenance is required by the project rule. A score from 40% to below 70% is classified as WARNING and indicates that maintenance should be scheduled soon. A score below 40% is classified as CRITICAL and indicates immediate maintenance is required.

X. MODEL EVALUATION

The model is evaluated using accuracy, a classification report, and a confusion matrix. The classification report provides precision, recall, and F1-score, while the confusion matrix compares actual and predicted classes and helps identify correct predictions, false alarms, and missed failures.

Actual Colab accuracy: _____ % (to be filled from the completed Colab notebook).

XI. FEATURE IMPORTANCE AND VISUALIZATION

Random Forest provides `feature_importances_` values that indicate the relative importance of the input features for the trained model. The project displays these values in descending order and visualizes them using a horizontal bar chart. The actual top-five features and their importance values should be inserted from the final Colab output.

Planned visualizations include vehicle health score for the first 30 test samples, failure probability for the first 30 test samples, vehicle status distribution, and feature-importance visualization.

XII. SAMPLE OUTPUT

The project specification gives the following sample output: Vehicle Health = 72%, Failure Probability = 68%, Status = WARNING, with major contributing factors listed as high vibration, rising temperature, and abnormal oil pressure. These are sample values and are not claims about an actual vehicle prediction.

XIII. ADVANTAGES AND LIMITATIONS

The prototype provides an automated vehicle health monitoring workflow, produces failure probability rather than only a binary

prediction, converts model output into an interpretable health score, generates maintenance alerts using predefined thresholds, and provides feature-importance information.

Limitations include the use of a generated prototype dataset rather than real vehicle telemetry, absence of live sensor connectivity, simple project-defined health thresholds, need for domain-specific validation of failure labels and patterns, and possible changes in model performance when authentic data from different vehicles or operating environments are used.

XIV. FUTURE SCOPE

Future work includes connecting the system to real-time vehicle sensors, using larger authentic vehicle maintenance datasets, comparing Random Forest with XGBoost and other suitable models, exploring LSTM or GRU models for sequential sensor data, adding explainable-AI methods, developing a web or mobile dashboard, and adding notification services for maintenance alerts.

XV. CONCLUSION

This project demonstrates an end-to-end AI/ML workflow for predictive maintenance and vehicle health monitoring. Vehicle operating parameters are preprocessed and supplied to a Random Forest classifier. The model output is transformed into failure probability and a vehicle health score, followed by a simple status and maintenance-alert mechanism. Visualization and feature importance provide additional understanding of model output. The prototype fulfills the main workflow described in the project specification and provides a foundation that can later be extended to real-time vehicle data.

REFERENCES

- [1] Module 3 (Connectivity) Project specification provided by the course/institution.
- [2] Scikit-learn documentation, including RandomForestClassifier, train_test_split, StandardScaler, and evaluation metrics.
- [3] Pandas documentation, CSV data handling and DataFrame operations.
- [4] Matplotlib and Seaborn documentation, data visualization.